```
SELECT channel, SUM(orders) AS total_orders

FROM gold.daily_sales

GROUP BY channel;


SELECT *

FROM gold.daily_sales

LIMIT 10;


SELECT COUNT(*) AS total_records

FROM gold.daily_sales;


SELECT *
FROM gold.daily_sales
ORDER BY month DESC
LIMIT 10;
```

- How many customers are in the database?
- Show the top 10 products by total sales revenue.
- What is the total sales revenue by month?
- Which 5 customers have placed the most orders?
- Are there any orders with missing customer IDs or invalid amounts?

AI Data Engineering & Analytics Copilot

Good morning, everyone.

I am Abid, and today I am going to present my project ppt on, **AI Data Engineering and Analytics Copilot**.

This project is a system designed to connect data discovery, data quality, data transformation and analytics within a controlled workflow.

In a typical data environment, engineers and analysts spend considerable time understanding datasets, identifying data-quality issues, investigating pipeline failures and writing queries to answer business questions.Our project aims to simplify these activities by coordinating specialized AI agents while maintaining security, authorization and human oversight.

Slide 2: The Problem

In many organizations, data is distributed across different files, databases and other sources. Before analysts can use it, they need to understand the available datasets, identify their schemas and check whether the data is reliable.

The second challenge is data quality. Missing values, duplicate records, inconsistent formats and other issues can affect analytical results. Investigating these problems and understanding pipeline failures can take significant time.

The third challenge is analytics. Business users may ask questions in simple English, but answering them can require knowledge of database structures and SQL.

Finally, introducing AI into these workflows creates a security challenge. We cannot allow an AI model to execute every operation without checking permissions or validating its actions.

Therefore, the main objective of this project is to connect discovery, quality, transformation and analytics in one controlled workflow.

Slide 3: System Architecture

This slide presents the high-level architecture of the system.We can understand it through three main layers.

The first is the **user-experience layer**. We use a multipage Streamlit interface through which users interact with the application, navigate its features and submit requests.

The second is the **intelligence layer**. The FastAPI backend receives requests, and LangGraph coordinates the workflow between specialized agents. These agents perform tasks such as metadata discovery, SQL planning, data-quality interpretation and ETL recommendations.

The system also connects with approved tools and data sources, including PostgreSQL, business-data sources and PySpark pipelines, as represented in the architecture.

The third is the **control layer**. This includes authentication, role-based access control, audit logging, approvals and monitoring.

These controls are important because an AI agent should not automatically receive unrestricted access to databases or operational tools.

Slide 4: Data Processing — Bronze, Silver and Gold

Now let us look at how data moves through the processing workflow.

The process starts with **ingestion**, where data is brought in from the available sources. The system also discovers relevant schemas and metadata to understand the incoming data.

The first layer is Bronze. Here, raw data is retained so that the original input remains available for traceability.

The second layer is Silver. This is where data-quality operations such as cleaning, validation and deduplication are performed.

For example, if a dataset contains duplicate records or inconsistent values, the Silver layer is where the applicable cleaning and validation rules can address those issues.

The third layer is Gold. Data is aggregated and prepared into a format suitable for reporting and analytics.

Quality checks and drift detection complement these stages. Monitoring also helps track pipeline status, failures and retries.

The main benefit of this structure is that raw data, cleaned data and analytics-ready data are organized into distinct stages.

This makes the data-processing workflow easier to understand and helps preserve traceability from source data to analytical outputs.

Slide 5: Agent Orchestration

Instead of treating the system as one general-purpose AI agent, we organize responsibilities into specialized tasks.

The first agent is **Metadata Discovery**. It identifies relevant datasets, schemas and columns.

The second is **SQL Planning**. It translates a natural-language question into an authorized query plan.

The third is **Data Quality**. It interprets quality-check results and explains the issues identified.

The fourth is **ETL Recommendation**. It explains relevant transformation approaches and possible pipeline improvements.

LangGraph coordinates these activities through shared state, allowing information produced during one stage to be used by subsequent stages.

Consider this business question: “Which five products generated the highest revenue?”

The workflow can first identify the relevant dataset and columns. It then prepares a query plan, validates the proposed operation, checks authorization, executes the permitted query and explains the resulting output.

Not every request needs to invoke every agent. The workflow depends on the task being performed.

The key point is that specialized agents contribute to a coordinated process rather than operating as completely independent systems.

Slide 6: Governance and Security

This is one of the most important parts of the project.

AI-generated operations must be controlled because an incorrect or unauthorized query can expose data or affect a system.

The workflow follows six stages: request, authorize, validate, approve, execute and audit.

First, the backend authenticates the user and checks their permissions through role-based access control.

Next, the proposed query or operation is validated against the applicable rules. Where configured, SQL abstract syntax tree restrictions help identify disallowed query structures.

Human approval is required when the workflow's rules call for it. Only then can the permitted operation proceed to execution.

The system also uses environment variables to protect sensitive configuration and secrets.

After execution, the control plane records information such as the actor, time and outcome for auditing.

For analytical operations, read-only access and query restrictions can reduce the risk of unintended changes.

The central principle is that **the language model may assist with planning, but it must not independently decide what it is authorized to execute**.

These controls establish a boundary between AI assistance and governed system operations.

Slide 7: End-to-End Demonstration

Now I will demonstrate the application's workflow.

Step 1: Login and application access

This establishes the starting point for interacting with the system. Authentication and permissions determine which operations the user is allowed to access.

Step 2: Dataset exploration and data quality

The purpose of this step is to understand what data is available and identify any reported quality issues before using it for analytics.

Step 3: Pipeline status and history

This helps us understand the operational state of the data-processing workflow and identify any reported failures or retries.

Step 4: Natural-language analytics

For example, I can ask: “Which five products generated the highest revenue?”

The system should identify the relevant data, prepare the query plan and apply the configured validation and authorization controls.

Step 5: Review the query or result

This step is important because users should be able to review what the system proposes or returns rather than treating every AI-generated response as automatically correct.

Step 6: Approval or blocked action

This shows how the system handles operations that should not proceed without the required authorization or validation.

Step 7: Logs and monitoring

These records help us understand what happened during the workflow and provide visibility into the outcome of operations.

This completes the demonstration path, from user access and data discovery through analytics, security controls and operational monitoring.

Slide 8: Conclusion and Thank You

To conclude, the AI Data Engineering and Analytics Copilot brings together data discovery, data-quality analysis, ETL workflows and natural-language analytics.